Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/21/2016
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	12/11/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chao, Beei-Huan Ph.D. (University of Hawaii at Manoa) Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the Principal Investigator, the three Co-Investigators, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland (UMD), and Beei-Huan Chao directing analytical work and computations at University of Hawaii.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	Summary of Accomplishments • The research team published 6 conference papers and abstracts during this period. • Work at Univ. of Maryland has started a new experimental study on the effects of dilution on the stoichiometric lengths of normal and inverse gas jet diffusion flames Two new graduate students were added to this project at UMD: Zhengyang Wang (Ph.D.) and Mohamed El Makawy (M.S.). They have generated predictions of these effects from the model of Roper (1977). Stoichiometric lengths of over 250 flames have been measured. Physical constants in the model of Roper have been adjusted such that it predicts these reasonably well. • Work at Washington University continued to pursue normal gravity experiments of soot formation in diffusion flames, numerical modeling, chemical kinetics, water-gas shift reaction rates, and laser-induced breakdown spectroscopy (LIBS). The LIBS measurement method has been improved. We now use a gated spectrometer to utilize C, H, N, O atomic lines instead of ionic lines in atomic ratio measurement. We improved accuracy with acoustic aid and automated baseline correction. Shot-to-shot variation is analyzed with relative standard deviation. • Work at Univ. of Hawaii continued to pursue analytical and computational studies of soot formation, radiation, and extinction in diffusion flames. Also being investigated are ignition behavior and effects of C/O ratio in diffusion flames. • The flight experiment was supported in various ways. For example, the research team responded to issues raised by NASA engineers in refining the test matrix and ACME requirements document. The research team also aided staff at NASA Glenn with burner characterization drop testing in the 2 s drop facility. Scientific Context, Hypothesis, and Research Objectives The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI, the three Co-Is, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, Peter Sunderland directing numerical simulations at University of Maryland, and Beei-Huan Chao directing analytical work and computations at University of Hawaii. Additional Accomplishments W. Wu, G. Yablonsky, R.L. Axelbaum, Observation of Water-Gas Shift Equilibrium in Diffusion Flames, Combustion and Flame, accepted. The kinetic behavior of the water-gas-shift (WGS) sub-system has been studied computationally for hydrocarbon diffusion flames in the counterflow geometry. Of specific interest was the spatial domain in which the reactions in the WGS-complex exhibit characteristics that lead to species concentrations approaching those expected from equilibrium. Two scenarios were observed for WGS equilibrium: (1) Full equilibrium in which every step of the WGS-mechanism is nearly in equilibrium; (2) “Apparent” equilibrium, which is achieved via opposing but coherent shifts away from equilibrium for the two key elementary reactions that constitute the WGS reaction. This interesting phenomenon is caused by the supply of the H radical, which participates in the forward reaction of one step and the reverse reaction of the other. This new form of apparent equilibrium differs from well-known “quasi” equilibrium, which is typically caused by the combination of a fast reversible step and slow irreversible step. Parametric domains where found in which these two scenarios are observed. For a given fuel type it was shown that the two scenarios can be separated by the location of stoichiometry. On the oxidizer side, full equilibrium of the overall WGS reaction is achieved, while on the fuel side, apparent equilibrium can occur. Z. Wang, M. El Makawy, P.B. Sunderland, B. Chao, R.L. Axelbaum, The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames, Abstract, ASGSR, Cleveland OH (2016). Many past studies have examined the stoichiometric lengths of laminar gas jet diffusion flames. These are found to be independent of pressure and burner diameter for fixed fuel mass flow rate. However, some studies report flame lengths increase when inert is added to the

fuel stream, whereas others report no change. Furthermore, measurements of effects of inert in the ambient gas are sparse, as are measurements for inverse flames. The analytical model of Roper (CNF 29:219, 1977) accounts for the full range of reactant dilution and inverse flames, but has only been tested for fuel-air systems.

This work examines the stoichiometric lengths of normal and inverse steady methane/oxygen/nitrogen jet flames burning on a 3.0 mm coflowing burner inside a glass chimney. Nitrogen dilution is such that methane mole fractions vary from 0.15 – 1 and oxygen mole fractions vary from 0.18 – 1. The measured flame lengths vary from 3 – 53 mm. Stoichiometric lengths are measured from visible emissions of the blue flame sheets. Reactant flow rates are measured with calibrated rotameters.

The predictions of the Roper model are compared with the measured flame lengths. For normal flames the agreement is best for a mass diffusivity at 25 C of 17 mm2/s and a temperature of 0.94 times the adiabatic flame temperature. This contrasts to values used by Roper for methane/air flames of 20 mm2/s and 1500 K, respectively. For the present conditions the predictions match the measurements with a R2 coefficient of determination of 0.91.

For fixed fuel flow rate, inert addition to the fuel stream is found to increase flame length, but this effect is negligible for fuel mole fractions above 0.5.

A simplified approximation of the Roper model has been obtained for normal methane/nitrogen/oxygen diffusion flames. This removes the requirement to specify adiabatic flame temperature and stoichiometric mixture fraction. The simplified model predicts flame length from the methane flow rate and mole fraction and the oxygen mole fraction.

W. Wu, R.L. Axelbaum, Interpreting Flame Structure by Mixture Fraction Measurement Using Laser-Induced Breakdown Spectroscopy, Abstract, ASGSR, Cleveland OH (2016). Measuring mixture fractions in a broad range of combustion systems with a suitable spatial and temporal resolution is critical not only for understanding the flame behavior, but also for the validation of rapidly developing numerical models. Instead of deriving the mixture fraction from measured molecular profile, a new approach of obtaining such quantities from atomic ratios measured by laser induced breakdown spectroscopy (LIBS) is introduced in this study. The atomic ratios for C, N, H, and O (e.g., C/O, H/O, and N/O) for the LIBS system were first calibrated in an ethylene-air premixed flat flame, where the equivalence ratio of the pre-mixture was varied from 0.60 to 1.80. Then the atomic ratio distributions along the centerline of an ethylene-air counter-flow diffusion flame were measured to demonstrate the capability of LIBS to measure atomic ratios between different elements throughout the entire flame. The elemental mass fractions for C, H, N, O can be thus obtained from atomic ratio profiles. The mixture fraction distribution was calculated from elemental mass fractions. Numerical results were compared to the experimental results and showed excellent agreement. The results of this work demonstrate that the LIBS technique holds promise as a simple tool for measuring atomic ratios and mixture fraction in diffusion flames.

Practical Applications. This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.

Plans for Upcoming Year Including Use of NASA Facilities. In the coming year we plan to continue with our publications of experimental and numerical work. We plan to perform additional simulations at Washington Univ., Univ. of Hawaii, and Univ. of Maryland. We plan to continue efforts to advance to flight tests and spherical burner development.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
Abstracts for Journals and Proceedings	Wang Z, El Makawy M, Sunderland PB, Chao B, Axelbaum RL. "The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames" 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, Ohio, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, Ohio, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Wu W, Axelbaum RL. "Interpreting Flame Structure by Mixture Fraction Measurement Using Laser-Induced Breakdown Spectroscopy," 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, Ohio, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, Ohio, October 26-29, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Wu W, Yablonsky G, Axelbaum RL. "Observation of water–gas shift equilibrium in diffusion flames." Combustion and Flame. 2016 Nov;173:57-64. http://dx.doi.org/10.1016/j.combustflame.2016.07.025 , Nov-2016
Papers from Meeting Proceedings	Wu W, Axelbaum RL. "Interpreting Flame Structure by Conserved Scalar Measurement Using Laser-Induced Breakdown Spectroscopy." Combustion Institute Central States Meeting, Knoxville, TN, May 15-17, 2016. Combustion Institute Central States Meeting, Knoxville, TN, May 15-17, 2016. , May-2016